Key relationships

Modern electrical machines

Calculating torque

Torque is the turning effect of a **force**.

The balance always measures force. This can be related to mass. Force and mass are linked by:

Force = mass x acceleration or **F** = **m** x **a**,

where **a** is usually the acceleration due to gravity, 9.81 m /s^2.

This allows us to calculate the force pressing down on the balance, generated by the dynamometer.

Torque = force x distance from the axis of rotation. It is measured in newton-metres (Nm).

Combining these equations:

Torque, **T** = **m** x **a** x **r**

where, here, **r** is the distance of the pressure point from the centre of the machine.

Putting in the value of **r** for the dynamometer used in this kit:

torque (in Nm) = **balance reading** (in kg) x **9.81** (m/s^2) x **0.03812** (m).

Calculating power

Power is the rate at which energy is delivered and is measured in watts (W).

For a rotating body:

Mechanical power = (2 x **n** x π x **T**) / 60 where **n** is the speed of the motor in r.p.m.

For a DC motor:

Electrical power supplied to the motor = $\mathbf{V_{IN}}$ x $\mathbf{I_{IN}}$

where $\mathbf{V_{IN}}$ = input voltage and $\mathbf{I_{IN}}$ = input current.

For a single phase AC motor:

Electrical power supplied to the motor = $\mathbf{V_{IN}}$ x $\mathbf{I_{IN}}$ x cos(**Θ**)

where $\mathbf{V_{IN}}$ is the AC rms input voltage, $\mathbf{I_{IN}}$ is the rms input current

and cos(**Θ**) is the power factor where **Θ** is the phase angle between $\mathbf{V_{IN}}$ and $\mathbf{I_{IN}}$.

For a three phase AC induction motor:

Electrical power supplied to the motor = 3 x electrical power supplied to one phase

Calculating efficiency

Efficiency (η) = (Mechanical power delivered / Electrical power supplied) x 100%

Calculating synchronous speed and slip

Synchronous speed in r.p.m = (f x 60) / (number of pairs of poles)

(Number of pairs of poles relates to how the motor windings are configured:

a 2-pole motor has one pair of poles, a 4-pole motor has two pairs of poles etc.)

Slip = ((synchronous speed - slip speed) / synchronous speed) x 100%

A three phase supply connected to the stator of an induction machine produces a magnetic field given by the formula:

$$Ns = (fs/p) \times 60$$

Where fs = supply frequency and p is the number of pole pairs - in our case 1.

Relationships between three phase star and delta systems

Three phase systems consist of machines connected in star and delta configuration. With U, V and W connections the line voltage is the voltage between any of U, V, W and ground. The phase voltage is the voltage between any two of U, V, W. The line and phase voltage are linked by the equation:

$$Vline = SQRT(3) \times Vphase$$